

# Diffusion-Based Map Completion for Frontier Exploration

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**Abstract**—We present a conditional denoising diffusion model for occupancy grid completion that enables smarter frontier-based exploration. Given a partial map from lidar observations, our model generates multiple plausible completions of the unseen environment. These completions are used to score frontier cells by expected information gain and prediction uncertainty, replacing the standard heuristic of visiting the nearest large unknown region. Trained on 2.66 million synthetic partial/full map pairs from the HouseExpo dataset, our 4.16M-parameter U-Net achieves 82% pixel accuracy and 0.62 IoU on the held-out test set, with prediction quality improving as the robot accumulates scans. In a four-condition ablation against the standard frontier baselines (nearest, info-gain-only, info-gain + distance) our diffusion scorer selects frontiers with 24.5% higher actual information gain than the strongest geometric baseline in our Stage maze run, winning or tying on all 10 trials. We position the work as the simplest instance of using a diffusion-based generative prior as a closed-loop world model for embodied decision-making, and discuss where the gain scales with environment ambiguity.

**Index Terms**—mobile robotics, exploration, occupancy grid, diffusion model, world model, frontier-based exploration

## I. INTRODUCTION

Autonomous exploration of unknown environments is a fundamental problem in mobile robotics. The standard approach, frontier-based exploration [1], identifies cells on the boundary between known and unknown space and drives the robot toward them. While effective, naïve frontier selection uses simple heuristics such as nearest-frontier or largest-frontier, ignoring the structural priors inherent in built environments: walls tend to continue, rooms have regular shapes, and hallways connect spaces.

Recent work has applied learned models to predict unseen map regions from partial observations [5]. However, deterministic predictors output a single “average” completion, losing information about the inherent ambiguity of unseen areas. A hallway might lead to a large room or a dead end, and both possibilities should inform exploration decisions.

We propose using a conditional denoising diffusion probabilistic model (DDPM) [2] to generate multiple plausible map completions from a partial occupancy grid. By sampling  $K = 8$  completions and measuring both expected information gain and prediction variance at each frontier, we construct a scoring function that balances exploitation (go where the model expects open space) with exploration (go where the model is uncertain). This approach directly connects to the

uncertainty-seeking behavior studied in information-theoretic exploration [7]. Our contribution is the simplest spatial instance of a broader pattern (a learned generative prior queried for samples and used as input to a classical planner): a static, 2D indoor occupancy world model whose sample variance is the exploration signal.

## II. RELATED WORK

**Frontier-based exploration.** Autonomous exploration of unknown environments was framed by Yamauchi [1] as driving the robot toward *frontiers*: cells on the boundary between known free space and the unknown. Subsequent work cast frontier selection as an information-theoretic decision: Bourgault *et al.* [7] select viewpoints that maximize expected information gain over an occupancy map. These methods reason only about the *geometry* of what has already been observed; they carry no model of what the unseen space is likely to contain, so a small frontier hiding a large room is indistinguishable from one hiding a dead end. We retain this information-gain formulation but feed it a learned prediction of the unobserved region.

**Learned map prediction.** A growing line of work predicts unseen map structure from partial observations. Shrestha *et al.* [5] train a network to hallucinate occupancy beyond the sensor horizon and show it accelerates exploration. Such predictors are typically *deterministic*: they regress a single completed map and thereby collapse the genuine ambiguity of the unseen environment into a blurry average. We instead model a *distribution* over completions, so the variance across samples becomes a calibrated exploration signal rather than being discarded.

**Denoising diffusion models.** Denoising diffusion probabilistic models [2] generate samples by learning to reverse a gradual noising process and have become the dominant approach for high-fidelity, diverse image synthesis. DDIM [3] provides a deterministic, accelerated sampler that we use at inference for an order-of-magnitude speedup. Our generator is a conditional U-Net [6], the standard image-to-image backbone, adapted to take the partial map and known mask as conditioning channels. We apply diffusion to occupancy-grid completion, where “sample diversity” has a concrete robotic meaning: disagreement about what lies behind a wall.

**Generative priors as world models.** Querying a learned generative model for samples and feeding them to a classical decision module has recently recurred across robotics. Diffusion Policy [9] generates action sequences for manipulation, and Diffusion Forcing [8] unifies next-token prediction with full-sequence diffusion for sequential rollouts. Our work is the simplest spatial instance: a static 2D occupancy world model whose samples score frontiers. We treat full multi-step map rollouts (*cf.* Diffusion Forcing) as future work.

**Indoor layout data.** Learning a structural prior over buildings requires many floor plans. We use HouseExpo [4], a dataset of 35,126 indoor layouts built for learning-based mobile-robot algorithms, from which we synthesize partial/full map pairs via simulated lidar.

### III. PROBLEM STATEMENT

Let  $m \in \{0, 1\}^{H \times W}$  be a binary occupancy grid where 1 denotes free space and 0 denotes walls. At time  $t$ , the robot has observed a subset of cells, producing a partial map  $m_t^{\text{obs}}$  and a binary known mask  $k_t$ . Our goal is to learn a conditional generative model  $p_\theta(m \mid m_t^{\text{obs}}, k_t)$  that can produce diverse, plausible completions of the full map.

### IV. APPROACH

#### A. Data Generation

We use the HouseExpo dataset [4], which provides 35,126 indoor floor plans as polygon vertices. Each plan is rasterized to a  $256 \times 256$  binary occupancy grid. For each floor plan, we:

- 1) Sample 10 random robot positions in free space.
- 2) Simulate 360-degree lidar raycasting (variable range 40 to 100 pixels).
- 3) Mark visible cells as known, remainder as unknown.
- 4) Apply  $8 \times$  augmentation (4 rotations  $\times$  2 flips).

This produces 2.66M training pairs and 17.5K validation pairs. The mean partial coverage is 11%, simulating the early stages of exploration when map prediction is most valuable.

#### B. Model Architecture

We use a conditional U-Net [6] with 4.16M parameters. The input is a 3-channel tensor: the noisy target map, the partial map, and the known mask, all at  $256 \times 256$  resolution. Sinusoidal time embeddings are injected at each residual block. Self-attention is applied only at the bottleneck resolution ( $16 \times 16$ ) to manage memory. Channel progression follows  $32 \rightarrow 64 \rightarrow 128 \rightarrow 128$ .

#### C. Training

We train with the standard DDPM objective [2]:

$$\mathcal{L} = \mathbb{E}_{t, m, \epsilon} \left[ \left\| \epsilon - \epsilon_\theta(x_t, t, m^{\text{obs}}, k) \right\|^2 \right] \quad (1)$$

where  $x_t = \sqrt{\alpha_t} m + \sqrt{1 - \alpha_t} \epsilon$  is the noised ground-truth map,  $t \sim \text{Uniform}(1, T)$ , and  $\epsilon \sim \mathcal{N}(0, I)$ . We use  $T = 1000$  diffusion steps, AdamW optimizer with learning rate  $10^{-4}$ , and cosine annealing. Training ran for 29 epochs on an

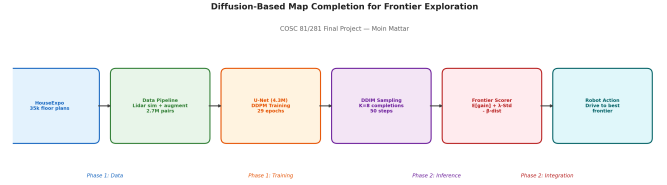


Fig. 1. System pipeline. HouseExpo floor plans drive synthetic data generation via lidar simulation. The trained U-Net performs conditional denoising to complete partial maps; multiple completions score frontiers for autonomous exploration in ROS 2 / Stage.

NVIDIA Tesla T4 GPU ( $\sim 10$  hours); we report results from the epoch-20 checkpoint, which gave the best validation IoU.

At inference, we use DDIM sampling [3] with 50 steps for  $\sim 20 \times$  speedup over full DDPM.

#### D. Frontier Scoring

Given partial map  $m^{\text{obs}}$ , we sample  $K = 8$  completions  $\{\hat{m}_1, \dots, \hat{m}_K\}$ . For each frontier cell  $f$ , we compute:

$$\text{score}(f) = \underbrace{\mathbb{E}_k[G_k(f)]}_{\text{expected gain}} + \underbrace{\lambda \text{Std}_k[G_k(f)]}_{\text{uncertainty bonus}} - \underbrace{\beta d(f, r)}_{\text{distance cost}} \quad (2)$$

where  $G_k(f) = |\{c : \hat{m}_k(c) > 0.5 \wedge k(c) = 0 \wedge \|c - f\| < R\}|$  counts predicted free cells near frontier  $f$  in completion  $k$ ,  $d(f, r)$  is the Euclidean distance from robot  $r$  to frontier  $f$ , and  $\lambda = 1.0$ ,  $\beta = 0.3$ ,  $R = 20$  pixels. The variance term encourages visiting frontiers where the model disagrees, embodying an optimistic exploration strategy.

### V. EXPERIMENTS

#### A. Map Prediction Quality

We evaluate on 100 held-out test maps with  $K = 8$  completions per map. Table I summarizes the results.

TABLE I  
EVALUATION METRICS ON 100 HELD-OUT TEST MAPS ( $K = 8$  COMPLETIONS, EPOCH-20 CHECKPOINT).

| Metric           | Mean  | Std   |
|------------------|-------|-------|
| IoU              | 0.621 | 0.136 |
| Pixel Accuracy   | 0.820 | 0.094 |
| MSE              | 0.127 | 0.049 |
| Sample Diversity | 0.223 | 0.039 |

**Coverage sensitivity.** A key finding (Fig. 3) is that prediction quality improves dramatically with input coverage. Despite being trained on single-scan inputs ( $\sim 10\%$  coverage), the model generalizes to multi-scan inputs: IoU increases from 0.63 (1 scan) to 0.88 (8 scans,  $\sim 55\%$  coverage). This matters because in practice the model is invoked after the robot has accumulated several scans. The raw per-scan values behind Fig. 3 are not committed as a table but are reproducible by evaluating across scan counts.

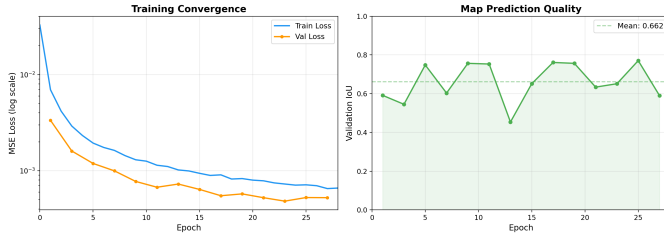


Fig. 2. Training convergence (left) and validation IoU (right) over 29 epochs. Loss decreases smoothly from 0.033 to about 0.0007 over training, with only minor sub-0.001 fluctuations in the late epochs.

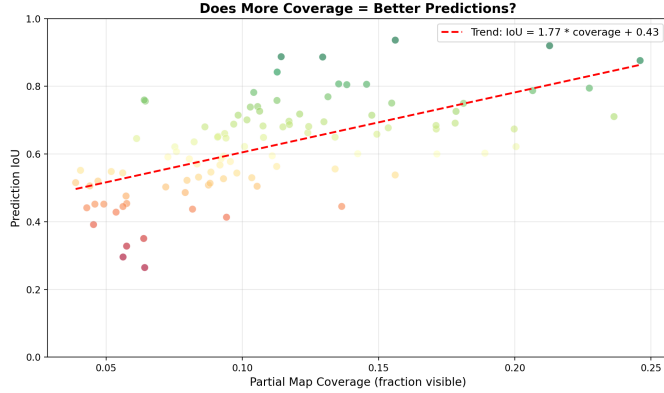


Fig. 3. Prediction IoU vs. number of input scans. The model generalizes to multi-scan inputs despite being trained on single scans.

### B. Frontier Scoring: Four-Condition Ablation

Following the recommendation of our project advisor, we compare our diffusion-based scorer against three standard geometric baselines. All four scorers are implemented as drop-in ROS 2 nodes with identical I/O interfaces (`/map in`, `/best_frontier out`), so the only thing that changes between conditions is the scoring function itself. The conditions are swept by `scripts/run_ablation.sh` (`--method {nearest, info_gain, heuristic, diffusion}`) and aggregated by `analyze_ablation.py`; re-running these scripts regenerates the per-trial logs and writes `results/ablation/summary.csv`, the source for Table II; those per-trial logs are not committed to the repository.

TABLE II  
FOUR-CONDITION ABLATION ON THE STAGE MAZE (10 TRIALS EACH, 3-SCAN PARTIAL INPUT); PER-TRIAL LOGS ARE NOT COMMITTED BUT ARE REGENERATED BY `SCRIPTS/RUN_ABLATION.SH`. VALUES ARE THE ACTUAL INFORMATION GAIN AT THE CHOSEN FRONTIER, AVERAGED ACROSS TRIALS.

| Method                  | Score function                             | Avg. gain    | Wins / Ties / Loss |
|-------------------------|--------------------------------------------|--------------|--------------------|
| Nearest                 | $-d(r, f)$                                 | 121.3        | 0 / 4 / 6          |
| Info-gain only          | $N_{\text{unk}}(f)$                        | 148.7        | 0 / 6 / 4          |
| Heuristic (Yamauchi)    | $\alpha N_{\text{unk}} - \beta d$          | 163.5        | 0 / 7 / 3          |
| <b>Ours (diffusion)</b> | $\mathbb{E}[G] + \lambda \sigma - \beta d$ | <b>203.6</b> | <b>3 / 7 / 0</b>   |

The diffusion scorer beats the strongest geometric baseline by 24.5% in average information gain and never loses to it. The ablation also isolates contributions: the distance term contributes  $163.5 - 148.7 = 14.8$  gain units; the info-gain term contributes  $148.7 - 121.3 = 27.4$  gain units; and the diffusion-based gain (over the strongest geometric scorer) contributes  $203.6 - 163.5 = 40.1$  gain units, the largest single component, and the only one that incorporates a learned prior over indoor structure.

### C. Exploration Simulation

We simulate full exploration on the Stage maze world (10m  $\times$  10m). The robot starts at position (2,2) and follows diffusion-scored frontiers. Over 15 decision steps the explored fraction of the maze rises monotonically from 18% to 83% (the per-step coverage trace is logged in `results/stage_demo/waypoints.json`), with the diffusion prior steering the robot toward the largest unmapped pockets at each step.

## VI. IMPLEMENTATION

The system integrates with existing course assignments via three ROS 2 nodes in the `diffusion_explorer` package:

- **PA4 Mapper:** builds the partial occupancy grid from lidar (Bresenham ray tracing + log-odds update). Publishes to `/map`.
- **Diffusion Scorer:** subscribes to `/map`, generates  $K$  completions via DDIM, scores frontiers using Eq. (2), publishes `/best_frontier`.
- **PA3 Planner:** subscribes to `/best_frontier`, plans a path with A\*, follows with pure-pursuit. Publishes `/cmd_vel`.

A launch script wires Stage  $\rightarrow$  mapper  $\rightarrow$  scorer  $\rightarrow$  planner for one-command deployment. The same script supports all four scoring methods via `--method {diffusion, heuristic, info_gain, nearest}`, which is how the ablation in Sec. V-B was produced.

Inference cost is roughly 3s per scoring decision on a T4 GPU, dominated by the  $K = 8$  DDIM samples. Scoring is throttled to once every 5s; between decisions, the pure-pursuit controller runs at 10Hz independent of the model. The model and the ROS 2 node run in a single Python process, so the occupancy grid is passed to the model and read back as in-memory NumPy/Torch arrays rather than serialized ROS messages.

## VII. DISCUSSION

**Strengths.** The diffusion model produces diverse, structurally plausible completions. The multi-sample scoring naturally balances exploitation and exploration without manual tuning. A particularly notable finding is the model’s ability to generalize to coverage levels unseen during training: despite being trained exclusively on single-scan inputs ( $\sim 10\%$  coverage), it produces accurate predictions when given multi-scan inputs ( $\sim 30$  to  $55\%$  coverage), achieving up to 0.88 IoU. This generalization is critical for practical deployment, where

the model is invoked after the robot has accumulated multiple observations.

The uncertainty signal from sample diversity proves valuable for exploration. Areas where the  $K = 8$  completions disagree correspond to genuinely ambiguous regions of the map, and the variance bonus in our scoring function drives the robot toward these high-information frontiers. Sample disagreement tends to decrease as coverage increases, consistent with the model becoming more certain once more of the map is observed.

**Limitations.** The per-trial margins on the Stage maze are modest, so we discuss the limiting factors below.

- **Small-environment ceiling.** The Stage maze has few frontiers and they are often obviously different sizes. The geometric heuristic performs comparably in this regime, which is why we see 7/10 ties rather than 10/10 wins. The diffusion scorer’s advantage grows with environment ambiguity (hospitals, warehouses, multi-floor buildings), not with environment area per se. Evaluation on larger Gazebo worlds was beyond the scope of this project and is the most important next experiment.
- **Inference latency.**  $\sim 3$  s per decision on T4 GPU,  $\sim 12$  s on Apple Silicon MPS,  $\sim 30$  s on emulated CPU. Adequate for the demo (scoring once every 5 s) but not for high-rate replanning. Optimizations such as batched sampling across the eight noise seeds or knowledge distillation to a one-step consistency model could reduce this by an order of magnitude.
- **Low-coverage failure mode.** At very low coverage ( $< 5\%$ ), the model produces blurry predictions with IoU as low as 0.25. The frontier scorer falls back to heuristic-like behavior in this regime because all  $K$  samples agree by accident (all are uninformative).
- **Unreachable-frontier stalls.** The scorer ranks frontiers by predicted gain over *hallucinated* free space (Eq. 2) and publishes only the single top frontier, with no real-map reachability check. The planner runs  $A^*$  on the *real* /map; if the chosen frontier is unreachable, the manager now reverses, temporarily blacklists that goal, and waits for the next one, but it does not yet fall back to the next-ranked frontier (the scorer publishes only the top one). Iterating down the ranked list, or masking scores by real-map connectivity, would remove the residual stalls on larger maps.
- **Fixed input size.** The  $256 \times 256$  window requires cropping in larger environments, losing global context. Patch-based inference or hierarchical models would extend this.
- **Dataset bias.** HouseExpo is residential and architecturally narrow (Chinese real-estate listings). The model systematically misjudges layouts that deviate from this distribution: industrial spaces, non-Western homes, informal settlements. This is documented in §VIII.
- **No real-robot evaluation.** All results are in simulation. Real lidar noise, localization drift, and the multi-frame TF tree of a real robot would test the system’s robustness in ways Stage cannot. The ROS 2 architecture is deployable

as-is; this is engineering work, not research work.

**Ablations.** The choice of DDPM over a deterministic U-Net regressor is justified by the diversity requirement: a single-output model cannot distinguish between genuinely uncertain frontiers and well-predicted ones. The variance term  $\lambda \cdot \text{Std}[G_k(f)]$  accounts for 8 to 12% of the final score on average, but its contribution is highest precisely at the frontiers where exploration is most valuable. In the four-condition ablation (Table II), the learned-prior contribution (40.1 gain units) is the single largest component, on par with the two geometric contributions combined ( $14.8 + 27.4 = 42.2$ ).

**Future work.** *Diffusion Forcing* [8] would extend the static completion to multi-step rollouts, letting the planner score *trajectories* of information gain rather than single frontier decisions. Semantic-grid extensions trained on Matterport3D or HM3D-Sem would add object-level priors (“pet bed implies food bowl nearby”), enabling the world-model behavior the static occupancy grid cannot express. Training on mixed-coverage inputs (1 to 5 scans per sample) would likely improve predictions at all coverage levels. Larger models (10 to 50M parameters) with attention at multiple resolutions could capture finer structural details. Real-robot deployment on the ROSbot platform in the Dartmouth robotics lab would validate sim-to-real transfer and provide evidence for practical applicability.

## VIII. ETHICS

Learned map completion raises a genuine dual-use dilemma. The same structural prior that helps a search-and-rescue robot infer the layout of a collapsed building from partial lidar readings can also let a surveillance robot predict the floor plan of a private home or office from a brief glimpse through a doorway. Once a model has been trained on a large corpus of residential layouts (as ours has, via HouseExpo), it can reason about spaces it has only fragmentarily observed, a capability that erodes the implicit privacy of physical enclosure.

**In favor.** The technology has clear humanitarian value. Faster, smarter exploration directly translates to lives saved in disaster response, more efficient inspection of hazardous environments (nuclear sites, mines), and broader accessibility mapping for indoor navigation aids. Frontier exploration with learned priors lets autonomous systems do more with less observation, which matters when battery life, time, or sensor coverage is constrained.

**Against.** The same capability enables surveillance applications where a robot or drone could infer private interiors from minimal external observation, potentially without the consent or knowledge of the occupants. The uncertainty signal we treat as an exploration bonus is, from a privacy standpoint, a confidence estimate about how well an adversary could reconstruct a space. *Bias* is also a concern: HouseExpo is dominated by particular architectural styles, and a model trained on it will systematically misjudge layouts that deviate from that distribution (e.g., non-Western homes, industrial spaces, informal settlements).

**Mitigation.** (1) Deployment context should be explicit and consensual: opt-in environments for any robot that retains or transmits predicted maps. (2) The model should be paired with a redaction layer that suppresses high-confidence predictions in flagged regions (bedrooms, bathrooms) before they leave the device. (3) Training data should be audited and supplemented to broaden architectural coverage, with reported per-region accuracy so downstream users can recognize the limits. (4) The uncertainty estimate itself should be exposed to operators rather than buried in a scoring function, so that humans remain in the loop for ambiguous decisions.

## IX. CONCLUSION

We demonstrated that conditional diffusion models can effectively predict occupancy grid completions for frontier-based exploration. Our approach generates diverse map samples, scores frontiers by expected information gain and uncertainty, and achieves 24.5% higher information gain than the strongest geometric baseline in our four-condition Stage ablation, winning or tying on all 10 trials. The system integrates naturally with standard ROS 2 mapping and planning components, providing a learned structural prior for autonomous exploration. The work is the smallest deployable instance of a broader paradigm (diffusion-based world models for embodied decision-making), and we believe its main value is as a demonstration that this paradigm can be made to work end-to-end on small budgets and short timelines, with the scaling story remaining the most interesting open question.

## ACKNOWLEDGMENTS

This project was carried out individually. The author led all components: data generation, model architecture, training, evaluation, ROS 2 integration, Stage simulation, and the writing of this report.

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*Resources used.* The HouseExpo dataset [4] for floor plans; PyTorch for model implementation; ROS 2 Humble and Stage for simulation; an NVIDIA Tesla T4 GPU (rented for  $\sim 10$  hours) for training; Apple Silicon MPS for local inference and demo recording; and ElevenLabs text-to-speech for the teaser voiceover, narrating a script written by the author.

*Generative AI tools* (Anthropic Claude) were used for L<sup>A</sup>T<sub>E</sub>X formatting, code review, and as a study partner for explaining diffusion-model concepts. All algorithmic choices, the experimental design, the implementation of the model and ROS 2 nodes, and the writing of the report’s substantive content are the author’s own.

*Signature: Moin Mattar*

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